

Cognevance Technologies

Java Full Stack Developer - Project Tasks

This document contains 3 industry-oriented projects for the **Java Full Stack Developer** domain. The projects are designed to help learners improve practical skills in Java backend development, frontend integration, databases, REST APIs, and enterprise application development.

Submission Instructions

- Create a separate GitHub repository for each project.
 - Repository naming format should be: **cognevance_projectName**.
 - Upload complete frontend and backend source code, screenshots, reports, and documentation.
 - Add a proper README.md file explaining setup process, technologies used, APIs, and workflow.
 - Deploy the projects using platforms such as Render, Railway, Netlify, or cloud servers whenever possible.
 - After completing all tasks, submit the GitHub repository link.
 - Send all repository links to **support@cognevance.online**.
 - Along with the submission, share your experience, review, and suggestions.
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Level 1 - Easy

Project Title: Student Management System

Project Overview: Build a basic full stack student management application using Java backend and frontend technologies.

Project Tasks:

- 1 Create responsive frontend pages using HTML, CSS, JavaScript, or React.
- 2 Develop backend APIs using Java Spring Boot.
- 3 Implement CRUD operations for student records.
- 4 Store data in MySQL database.
- 5 Create forms for adding, updating, and deleting students.

- 6 Deploy the application online.
- 7 Prepare screenshots and project documentation.

Expected Deliverables:

- Frontend source code
 - Backend source code
 - Database setup
 - Screenshots/demo
 - Deployment links
 - GitHub repository link
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Level 2 - Intermediate

Project Title: Online Banking Management System

Project Overview: Develop a secure full-stack banking application with authentication and transaction management.

Project Tasks:

- 1 Implement user authentication and authorization using Spring Security.
- 2 Develop REST APIs using Spring Boot.
- 3 Create modules for account management and transactions.
- 4 Integrate MySQL/PostgreSQL database.
- 5 Build responsive frontend dashboard.
- 6 Implement transaction history and account balance tracking.
- 7 Deploy the application and test APIs.

Expected Deliverables:

- Complete full stack project source code
- REST API documentation
- Database schema
- Screenshots/demo
- Deployment links
- GitHub repository link

Level 3 - Advanced

Project Title: Enterprise ERP & Business Analytics Platform

Project Overview: Build a scalable enterprise-level ERP and analytics platform using Java full stack technologies.

Project Tasks:

- 1 Develop secure authentication and role-based authorization system.
- 2 Create modules for employees, inventory, sales, and finance management.
- 3 Implement advanced REST APIs and backend microservices.
- 4 Integrate analytics dashboards and reporting systems.
- 5 Use cloud databases and file storage integration.
- 6 Optimize backend performance and application scalability.
- 7 Deploy the complete enterprise application.
- 8 Document architecture, APIs, and workflows.

Expected Deliverables:

- Complete enterprise application
- Frontend and backend source code
- Database schema
- Analytics dashboard
- Deployment links
- Project documentation/report
- GitHub repository link

Important Note:

Maintain clean code, professional documentation, and proper folder structure inside your GitHub repository.

For any support or queries, contact: **support@cognevance.online**